

## DUMBRAVITA DE CODRU, Romania

WMO: 151380

Lat: 46.6447N

Lon: 22.1711E

Elev: 588

StdP: 94.46

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 1             | -9.9       | -7.9 | -15.2                         | 1.1 | -6.2 | -12.4 | 1.4 | -5.0 | 11.8                  | 5.3  | 8.4 | 2.2  | 1.4                   | 160  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.2                    | 31.0            | 20.5 | 29.1 | 20.0 | 27.6 | 19.4 | 22.2                | 28.1 | 21.3 | 27.1 | 20.4 | 26.1 | 1.5                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.5                            | 16.3 | 25.0 | 19.4 | 15.2 | 24.4 | 18.4 | 14.3 | 23.7 | 68.5          | 28.1 | 64.7 | 27.2 | 61.4 | 26.3 | 26.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 8.2           | 6.2  | 5.0 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.0        | -0.5  | 2.4   | 5.4   | 10.9  | 15.7  | 19.3  | 20.6  | 21.9  | 16.4  | 11.5  | 6.9   | 1.1   |  |  |
|                                                                        | DBStd       | 8.88        | 4.92  | 4.65  | 4.47  | 5.73  | 4.11  | 3.75  | 3.60  | 4.21  | 5.06  | 4.98  | 5.45  | 4.61  |  |  |
|                                                                        | HDD10.0     | 1202        | 325   | 214   | 153   | 60    | 6     | 0     | 0     | 0     | 4     | 43    | 121   | 277   |  |  |
|                                                                        | HDD18.3     | 3011        | 583   | 446   | 399   | 228   | 103   | 33    | 17    | 15    | 96    | 216   | 342   | 533   |  |  |
|                                                                        | CDD10.0     | 1578        | 1     | 2     | 12    | 86    | 181   | 279   | 330   | 368   | 197   | 91    | 29    | 2     |  |  |
|                                                                        | CDD18.3     | 344         | 0     | 0     | 0     | 4     | 20    | 62    | 89    | 124   | 39    | 5     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2193        | 0     | 0     | 0     | 9     | 64    | 355   | 582   | 956   | 226   | 2     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 568         | 0     | 0     | 0     | 0     | 3     | 67    | 157   | 291   | 51    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.7         | 1.5   | 2.0   | 1.8   | 1.9   | 1.7   | 1.6   | 1.5   | 1.5   | 1.7   | 1.5   | 2.1   | 1.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 866         | 63    | 49    | 56    | 71    | 80    | 103   | 96    | 66    | 78    | 67    | 65    | 72    |  |  |
|                                                                        | PrecMax     | 1109        | 100   | 126   | 189   | 123   | 164   | 222   | 183   | 120   | 189   | 151   | 119   | 160   |  |  |
|                                                                        | PrecMin     | 590         | 28    | 10    | 15    | 20    | 34    | 28    | 36    | 12    | 16    | 3     | 6     | 10    |  |  |
|                                                                        | PrecStd     | 143         | 20    | 26    | 38    | 27    | 33    | 41    | 41    | 33    | 40    | 39    | 26    | 36    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 11.7  | 15.0  | 19.3  | 24.3  | 26.8  | 30.5  | 32.8  | 33.6  | 30.7  | 23.4  | 20.1  | 13.6  |  |  |
|                                                                        |             | MCWB        | 7.2   | 8.5   | 11.3  | 14.9  | 17.6  | 21.6  | 21.4  | 21.4  | 19.4  | 15.3  | 12.7  | 7.8   |  |  |
|                                                                        | 2%          | DB          | 10.0  | 12.3  | 16.2  | 22.5  | 25.2  | 28.6  | 30.2  | 31.5  | 27.8  | 21.6  | 17.2  | 11.5  |  |  |
|                                                                        |             | MCWB        | 6.5   | 7.8   | 9.6   | 13.6  | 17.5  | 20.9  | 20.7  | 20.3  | 17.8  | 14.2  | 11.4  | 7.0   |  |  |
|                                                                        | 5%          | DB          | 8.0   | 10.6  | 14.1  | 20.6  | 23.7  | 27.1  | 28.2  | 29.7  | 25.7  | 20.1  | 15.9  | 9.3   |  |  |
|                                                                        |             | MCWB        | 5.1   | 6.8   | 8.3   | 12.3  | 16.8  | 19.9  | 20.1  | 19.9  | 16.8  | 13.3  | 10.8  | 5.6   |  |  |
|                                                                        | 10%         | DB          | 6.5   | 9.0   | 12.0  | 18.6  | 22.0  | 25.5  | 26.4  | 27.9  | 23.9  | 18.5  | 14.0  | 7.5   |  |  |
|                                                                        |             | MCWB        | 4.2   | 5.6   | 7.2   | 11.4  | 16.0  | 19.2  | 19.2  | 19.3  | 16.3  | 12.9  | 9.5   | 4.8   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 8.1   | 10.0  | 12.1  | 16.2  | 19.9  | 23.3  | 23.5  | 24.0  | 20.7  | 17.3  | 13.4  | 8.7   |  |  |
|                                                                        |             | MCDB        | 10.4  | 13.5  | 18.0  | 22.8  | 24.1  | 29.0  | 28.9  | 28.2  | 28.5  | 21.5  | 18.2  | 12.1  |  |  |
|                                                                        | 2%          | WB          | 7.0   | 8.2   | 10.3  | 14.6  | 18.3  | 21.9  | 22.1  | 22.0  | 18.8  | 15.4  | 12.1  | 7.5   |  |  |
|                                                                        |             | MCDB        | 9.5   | 11.0  | 15.9  | 20.7  | 23.4  | 27.2  | 27.4  | 28.6  | 25.2  | 19.3  | 16.8  | 11.1  |  |  |
|                                                                        | 5%          | WB          | 5.9   | 7.2   | 8.8   | 13.3  | 17.2  | 20.8  | 21.1  | 21.0  | 17.7  | 14.3  | 11.2  | 6.2   |  |  |
|                                                                        |             | MCDB        | 7.7   | 10.0  | 13.3  | 19.0  | 22.7  | 26.2  | 26.5  | 27.6  | 24.1  | 18.7  | 15.2  | 8.8   |  |  |
|                                                                        | 10%         | WB          | 4.3   | 6.2   | 7.5   | 12.1  | 16.3  | 19.8  | 20.1  | 20.1  | 16.8  | 13.4  | 10.3  | 5.2   |  |  |
|                                                                        |             | MCDB        | 6.1   | 8.5   | 11.4  | 17.8  | 21.5  | 24.9  | 25.1  | 26.7  | 22.7  | 17.8  | 13.6  | 7.3   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.0   | 5.9   | 7.2   | 8.2   | 7.8   | 8.1   | 8.7   | 8.2   | 7.1   | 6.2   | 5.2   | 4.6   |  |  |
|                                                                        |             | MCDBR       | 6.2   | 7.5   | 9.1   | 9.4   | 9.0   | 9.3   | 10.0  | 9.1   | 8.7   | 7.5   | 6.5   | 5.8   |  |  |
|                                                                        | 5% WB       | MCWBR       | 4.1   | 4.5   | 5.0   | 5.2   | 5.8   | 5.4   | 5.4   | 4.5   | 4.2   | 4.3   | 3.6   | 3.6   |  |  |
|                                                                        |             | MCWBR       | 5.4   | 6.8   | 8.4   | 9.0   | 8.4   | 8.6   | 8.8   | 7.9   | 7.0   | 6.1   | 5.4   | 5.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.305 | 0.329 | 0.365 | 0.413 | 0.399 | 0.398 | 0.412 | 0.414 | 0.385 | 0.353 | 0.328 | 0.306 |  |  |
|                                                                        |             | taud        | 2.383 | 2.340 | 2.290 | 2.175 | 2.266 | 2.297 | 2.260 | 2.254 | 2.311 | 2.398 | 2.422 | 2.417 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 784   | 835   | 854   | 841   | 867   | 868   | 848   | 830   | 822   | 797   | 748   | 740   |  |  |
|                                                                        |             | Edh at Noon | 72    | 92    | 112   | 138   | 130   | 127   | 130   | 126   | 109   | 86    | 69    | 63    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.15  | 1.97  | 3.17  | 4.45  | 5.38  | 5.91  | 5.80  | 5.31  | 3.75  | 2.53  | 1.44  | 0.90  |  |  |
|                                                                        | RadStd      | RadStd      | 0.15  | 0.28  | 0.45  | 0.44  | 0.51  | 0.46  | 0.49  | 0.50  | 0.42  | 0.29  | 0.17  | 0.12  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (68 neighbors) |  | +0.70 | N/A     | N/A    | +1.00   | N/A   | N/A   | -127        | -191    | +112    | +57     |

Nomenclature: See separate page